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Least squares estimation techniques are developed for a special multiplicative model based on the Luce choice axiom whose potential usefulness in marketing applications justifies estimation techniques which can be easily implemented.

Parameter Estimation for a Multiplicative Competitive Interaction Model—Least Squares Approach

INTRODUCTION

In a recent attempt to measure geographic mobility of food and grocery shoppers, Haines, Simon, and Alexis [4] used a model initially developed by Huff [6]. The specification of the model is:

$$P_{ij} = \frac{\left(\dfrac{S_j}{T_{ij}^{\lambda}}\right)}{\left(\sum_{j=1}^{m} \dfrac{S_j}{T_{ij}^{\lambda}}\right)} \tag{1}$$

where:

S_j = the size of retail location j, $j = 1, 2, \ldots, m$;

T_{ij} = the distance separating a consumer at location i from retail location j, $i = 1, 2, \ldots, M$;

P_{ij} = the probability that a consumer living at i will choose to shop at j;

λ = sensitivity of spatial difference discrimination parameter (usually assumed to be greater than zero).

Haines et al. pointed out that this model is a special case of the Luce choice axiom [10] and may have broad applicability to consumer choice problems. In fact, a number of attempts have been made to apply extensions to the Huff model to analysis of competitive market behavior [8, 9, 13, 15]. A similar formulation has been used by Kuehn, McGuire, and Weiss [7] for measuring advertising effectiveness, and by Pessemier, Burger, Teach, and Tigert [14] for relating attitude scale values to brand purchase probability. A more general statement of this type of model may be given by:

$$\pi_{ij} = \frac{\prod_{k=1}^{q} x_{kij}^{\beta_k}}{\left(\sum_{j=1}^{m}\prod_{k=1}^{q} x_{kij}^{\beta_k}\right)} \tag{2}$$

where:

π_{ij} = the probability that a consumer in the ith choice situation selects the jth object;

x_{kij} = the kth variable describing object j in choice situation i;

β_k = the parameter for sensitivity of π_{ij} with respect to variable k.

It has been known that the above formulation has desirable properties as a model of brand share determination. Kotler [8] argues that a firm's marketing effectiveness is dependent on what its competitors do and this model captures the essence of competitive interactions. For this reason the model specified by (2) will be called the Multiplicative Competitive Interaction model or the MCI model for short. To be sure, the MCI model is not the only model which takes competitive interactions into account. Beckwith [1], for example, used multivariate linear regression models to measure advertising effects in a competitive

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environment. But the MCI model has an advantage that estimated market shares are guaranteed to be greater than zero and sum to unity, a condition which, as Naert and Bultez [12] pointed out, cannot be easily met by multivariate linear models.

The reason a model with such apparent marketing applications has not been more widely used seems to be the difficulty in parameter estimation. The nonlinear formulation of the model prompted past researchers to believe that it could not be estimated by standard econometric methods [9, p. 120] and various numerical estimation techniques have been developed. Urban [15] used an on-line iterative search program which utilized the decision maker's judgment as an input. Kuehn et al. used a nonlinear least squares method which minimized the sum of squared residuals by a direct search technique [7]. Hlavac and Little [5] and Haines et al. [4] attempted to find maximum likelihood estimates by Newton's method or a direct search technique.

All of the above techniques suffer from several common problems: (a) none guarantees that the global maxima or minima will be found; (b) they are all costly in terms of computing time; and (c) the statistical properties of the estimates are unknown, except for the asymptotic properties of maximum likelihood estimates. Haines et al. [4] show that maximum likelihood estimates—assuming the global maxima can be found—are consistent, but they are not the minimum variance estimates. Considering the potential usefulness of the MCI model, it seems desirable to have more practical parameter estimation techniques. This article will show that standard econometric techniques are indeed applicable to the MCI model and report the results of a simulation experiment to investigate the statistical properties of estimators.

PARAMETER ESTIMATION BY LEAST SQUARES TECHNIQUES

In order to develop least squares estimation techniques for the MCI model, it is necessary to transform (2) into a linear form with respect to the parameters of the model, i.e., the β_k's. Before any transformation can be performed, however, it must be pointed out that specification (2) is incomplete in the sense that it does not contain a disturbance term. When fitting this model to available data, such a term is necessary because of the possibility of model misspecification, i.e., the omission of some unknown or unavailable variables from the model. In the following development, specification errors are handled in the manner proposed by Bultez and Naert [2]. Redefine (2) as:

$$\pi_{ij} = \frac{\left(\prod_{k=1}^{q} x_{kij}^{\beta_k}\right)\zeta_{ij}^*}{\left[\sum_{j=1}^{m}\left(\prod_{k=1}^{q} x_{kij}^{\beta_k}\right)\zeta_{ij}^*\right]} \tag{3}$$

where ζ_{ij}^* is the "specification error" term. We assume that $\log \zeta_{ij}^*$ is independently and normally distributed with mean 0 and variance σ_ζ^2 (that is, ζ_{ij}^* is log-normally distributed). It has been shown in [13] that (3) can be written as a linear expression in the parameters, β_k.

First, take the logarithm of (3):

$$\log \pi_{ij} = \sum_{k=1}^{q} \beta_k \log x_{kij} + \log \zeta_{ij}^* - \log\left[\sum_{j=1}^{m}\left(\prod_{k=1}^{q} x_{kij}^{\beta_k}\right)\zeta_{ij}^*\right]. \tag{4}$$

Summing (4) over j (= 1, 2, ..., m) and dividing both sides by m, we have:

$$\frac{1}{m}\sum_{j=1}^{m}\log \pi_{ij} = \sum_{k=1}^{q}\beta_k\left(\frac{1}{m}\sum_{j=1}^{m}\log x_{kij}\right) + \frac{1}{m}\sum_{j=1}^{m}\log \zeta_{ij}^* - \log\left[\sum_{j=1}^{m}\left(\prod_{k=1}^{q} x_{kij}^{\beta_k}\right)\zeta_{ij}^*\right]. \tag{5}$$

If we let $\tilde{\pi}_i$, $\tilde{x}_{ki}$, and $\tilde{\zeta}_i^*$ be geometric means of π_{ij}, x_{kij}, and ζ_{ij}^* over i, i.e.,

$$\tilde{\pi}_i = \left(\prod_{j=1}^{m}\pi_{ij}\right)^{1/m}, \quad \tilde{x}_{ki} = \left(\prod_{j=1}^{m} x_{kij}\right)^{1/m}$$

$$\text{and } \tilde{\zeta}_i^* = \left(\prod_{j=1}^{m}\zeta_{ij}^*\right)^{1/m},$$

then (5) can be written as:

$$\log\left[\sum_{j=1}^{m}\left(\prod_{k=1}^{q} x_{kij}^{\beta_k}\right)\zeta_{ij}^*\right] = \sum_{k=1}^{q}\beta_k \log \tilde{x}_{ki} + \log \tilde{\zeta}_i^* - \log \tilde{\pi}_i. \tag{6}$$

The substitution of the right-hand side of (6) for the $\log[\Sigma_j(\Pi_k x_{kij}^{\beta_k})\zeta_{ij}^*]$ term in (4) yields, after rearranging terms,

$$\log\left(\frac{\pi_{ij}}{\tilde{\pi}_i}\right) = \sum_{k=1}^{q}\beta_k \log\left(\frac{x_{kij}}{\tilde{x}_{ki}}\right) + \log\left(\frac{\zeta_{ij}^*}{\tilde{\zeta}_i^*}\right). \tag{7}$$

Since (7) is linear in the β_k's, the use of multiple regression analysis is suggested to estimate the β_k's. But (7) cannot directly be treated as a regression model because true probabilities, π_{ij}, are not observable. Instead the observed proportions, p_{ij}, will have to be used as the dependent variable. If we define a new set of variables for each choice object j in choice situation i such that:

$$y_{ij} = \log\left(\frac{p_{ij}}{\tilde{p}_i}\right)$$

$$z_{kij} = \log\left(\frac{x_{kij}}{\tilde{x}_{ki}}\right),$$

where $\tilde{p}_i$ is the geometric mean of p_{ij} over j, then the relevant regression equation may be stated as:

$$y_{ij} = \sum_{k=1}^{q} \beta_k z_{kij} + \epsilon_{ij} \quad i = 1, 2, \ldots, M \quad j = 1, 2, \ldots, m \tag{8}$$

where ϵ_{ij} is the stochastic disturbance term. For later development it is convenient to state (8) in matrix notation. Let

$$\mathbf{y}_i' = (y_{i1}\ y_{i2} \cdots y_{im})$$

$$\mathbf{Z}_i = \begin{pmatrix} z_{1i1}\ z_{2i1} \cdots z_{qi1} \\ z_{1i2}\ z_{2i2} \cdots z_{qi2} \\ \cdots \\ z_{1im}\ z_{2im} \cdots z_{qim} \end{pmatrix}$$

$$\epsilon_i' = (\epsilon_{i1}\ \epsilon_{i2} \cdots \epsilon_{im})$$

and

$$\mathbf{y} = \begin{pmatrix} \mathbf{y}_1 \\ \hline \mathbf{y}_2 \\ \hline \vdots \\ \hline \mathbf{y}_M \end{pmatrix} \quad \mathbf{Z} = \begin{pmatrix} \mathbf{Z}_1 \\ \hline \mathbf{Z}_2 \\ \hline \vdots \\ \hline \mathbf{Z}_M \end{pmatrix} \quad \epsilon = \begin{pmatrix} \epsilon_1 \\ \hline \epsilon_2 \\ \hline \vdots \\ \hline \epsilon_M \end{pmatrix}.$$

Then (8) can be written as:

$$\mathbf{y} = \mathbf{Z}\beta + \epsilon \tag{9}$$

where $\beta' = (\beta_1\ \beta_2 \ldots \beta_q)$.

In applying multiple regression analysis to (9), it is necessary to consider the distribution characteristics of the error term, ϵ. Particularly important is the fact that there are two sources of errors which generate ϵ. Besides the "specification errors"—represented by ζ_{ij}^*—there is another type of errors which arise from the fact that the observed proportions p_{ij} are generated by a multinomial sampling process with underlying parameters, π_{ij}. This latter type of errors—"sampling errors"—is particularly relevant for sample survey data such as used by Haines et al. [4]. In order to give formal definitions to these two types of errors, decompose ϵ_{ij} into two parts as follows:

$$\begin{aligned} \epsilon_{ij} &= y_{ij} - \sum_{k=1}^{q} \beta_k z_{kij} \\ &= \left[y_{ij} - \log\left(\frac{\pi_{ij}}{\tilde{\pi}_i}\right)\right] + \left[\log\left(\frac{\pi_{ij}}{\tilde{\pi}_i}\right) + \sum_{k=1}^{q} \beta_k z_{kij}\right] = \eta_{ij} + \zeta_{ij} \end{aligned} \tag{10}$$

where:

$$\eta_{ij} = y_{ij} - \log\left(\frac{\pi_{ij}}{\tilde{\pi}_i}\right) = \log\left(\frac{p_{ij}}{\tilde{p}_i}\right) - \log\left(\frac{\pi_{ij}}{\tilde{\pi}_i}\right),$$

$$\zeta_{ij} = \log\left(\frac{\pi_{ij}}{\tilde{\pi}_i}\right) - \sum_{k=1}^{q} \beta_k z_{kij}.$$

Defined in this manner, η_{ij} results as the consequence of the observed proportions, p_{ij}, deviating from the true parameters, π_{ij}, and hence represents the sampling error term. ζ_{ij} represents the specification error term since, by (7),

$$\zeta_{ij} = \log\left(\frac{\zeta_{ij}^*}{\tilde{\zeta}_i^*}\right).$$

In order to examine the distribution characteristics of ϵ_{ij}, those of η_{ij} and ζ_{ij} are first examined. The specification errors, ζ_{ij}, are contemporaneously correlated for each choice situation i since:

$$\begin{aligned} \sum_{j=1}^{m} \zeta_{ij} &= \sum_{j=1}^{m} \log\left(\frac{\zeta_{ij}^*}{\tilde{\zeta}_i^*}\right) \\ &= \sum_{j=1}^{m} \left(\log \zeta_{ij}^* - \frac{1}{m}\sum_{j=1}^{m} \log \zeta_{ij}^*\right) = 0. \end{aligned}$$

If we let

$$\zeta_i' = (\zeta_{i1}\ \zeta_{i2} \cdots \zeta_{im})$$

$$\log \zeta_i^{*\prime} = (\log \zeta_{i1}^*\ \log \zeta_{i2}^* \ldots \log \zeta_{im}^*)$$

we may write:

$$\zeta_i = \left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right) \log \zeta_i^*$$

where $\mathbf{I}$ is the $m \times m$ identity matrix and $\mathbf{J}$ is an $m \times m$ summation matrix (i.e., all elements equal unity). By the assumption previously made, ζ_i is normally distributed with mean vector $\mathbf{0}$ and contemporaneous covariance matrix:

$$\Sigma_{\zeta i} = \left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right) \sigma_\zeta^2 \mathbf{I} \left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right) = \sigma_\zeta^2 \left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right). \tag{11}$$

Sampling errors result from the discrepancies between $y_{ij} = \log(p_{ij}/\tilde{p}_i)$ and $\log(\pi_{ij}/\tilde{\pi}_i)$ as defined by (10). Let

$$\eta_i' = (\eta_{i1}\ \eta_{i2} \cdots \eta_{im}).$$

Also let

$$\pi_i' = (\pi_{i1}\ \pi_{i2} \cdots \pi_{im})$$

$$\mathbf{p}_i' = (p_{i1}\ p_{i2} \cdots p_{im}).$$

Given a multinomial process with parameter vector π_i and a finite sample size n_i, the exact distribution of η_i is indeterminate because there is a probability greater than zero that $p_{ij} = 0$ (and consequently $\log p_{ij} = -\infty$) even if $\pi_{ij} > 0$. Fortunately, for a reasonably large sample size, the random vector $\mathbf{p}_i$ is approximately normally distributed around mean vector π_i.

Furthermore, most of the observations are concentrated in the relative neighborhood of π_i. Under these conditions, it can be shown that η_i is also approximately normally distributed with mean vector $\mathbf{0}$ and contemporaneous covariance matrix:

$$(12) \qquad \Sigma_{\eta i} = \frac{1}{n_i}\left[\Pi_i^{-1} - \frac{1}{m}(\mathbf{J}\Pi_i^{-1} + \Pi_i^{-1}\mathbf{J}) + c\mathbf{J}\right],$$

where $c = \Sigma_{j=1}^{m}(1/\pi_{ij})/m^2$, Π_i is an $m \times m$ diagonal matrix with diagonal entries of $(\pi_{i1}\ \pi_{i2} \ldots \pi_{im})$ and $\mathbf{J}$ is the summation matrix as defined before. See the Appendix for derivation of (12).

We now combine the results obtained so far. Since ζ_i is normally distributed by assumption, and η_i is approximately normally distributed for sufficiently large sample sizes, ϵ_i is approximately normally distributed with mean vector $\mathbf{0}$ and contemporaneous covariance matrix:

$$\Sigma_{\epsilon i} = \Sigma_{\eta i} + \Sigma_{\zeta i} + E\eta_i\zeta_i'$$

where $E\eta_i\zeta_i'$ is the matrix of covariances between η_{ij} and ζ_{ij}, i.e., its (j, l)th element is $\mathrm{cov}(\eta_{ij}, \zeta_{il})$. But for a reasonably large sample size,

$$\begin{aligned}\mathrm{cov}(\eta_{ij}, \zeta_{il}) &= E(\eta_{ij}\cdot\zeta_{il})\\ &= E[E(\eta_{ij}\mid\zeta_{il})\cdot\zeta_{il}] \cong 0\end{aligned}$$

since $E(\eta_{ij} \mid \zeta_{il}) \cong 0$ regardless of the value of ζ_{il}. Hence,

$$(13) \qquad \Sigma_{\epsilon i} \cong \Sigma_{\eta i} + \Sigma_{\zeta i}.$$

This is an important result, since it shows that the covariance matrix for ϵ_i is approximately equal to the sum of covariance matrices for its component vectors η_i and ζ_i even though η_i and ζ_i are not independently distributed.

Let us now examine the properties of the contemporaneous covariance matrix $\Sigma_{\epsilon i}$. If we let its (j,l)th element be σ_{jl}, by (11) and (12),

$$(14)\ \sigma_{jl} \cong \frac{m-1}{m}\sigma_\zeta^2 + \frac{1}{n_i}\left[\left(1-\frac{2}{m}\right)\left(\frac{1}{\pi_{ij}}\right) + c\right] \quad \text{if } j = l$$

$$\cong -\frac{1}{m}\sigma_\zeta^2 + \frac{1}{n_i}\left[-\frac{1}{m}\left(\frac{1}{\pi_{ij}} + \frac{1}{\pi_{il}}\right) + c\right] \quad \text{if } j \neq l$$

where $c = \Sigma_{j=1}^{m}(1/\pi_{ij})/m^2$. In general $\Sigma_{\epsilon i}$ is heteroscedastic and nonspherical. Under these conditions, the ordinary least squares (OLS) estimator:

$$\tilde{\beta}_o = (\mathbf{Z}'\mathbf{Z})^{-1}(\mathbf{Z}'\mathbf{y})$$

is not the minimum variance estimator, although it is still unbiased and consistent [3, p. 234]. The situation clearly calls for the use of a generalized least squares (GLS) estimator. Let Ω be the covariance matrix for ϵ, i.e., $\Omega = E\,\epsilon\epsilon'$ which is a block-diagonal matrix of the following form:

$$\begin{pmatrix} \Sigma_{\epsilon 1} & \mathbf{0} & \ldots & \mathbf{0} \\ \mathbf{0} & \Sigma_{\epsilon 2} & \ldots & \mathbf{0} \\ \ldots & \ldots & \ldots & \ldots \\ \mathbf{0} & \mathbf{0} & \ldots & \Sigma_{\epsilon M} \end{pmatrix}$$

Then the GLS estimator, given by

$$(15) \qquad \hat{\beta}_g = (\mathbf{Z}'\Omega^{-1}\mathbf{Z})^{-1}(\mathbf{Z}'\Omega^{-1}\mathbf{y})$$
$$= \left[\sum_{i=1}^{M}(\mathbf{Z}_i'\Sigma_{\epsilon i}^{-1}\mathbf{Z}_i)\right]^{-1}\left[\sum_{i=1}^{M}(\mathbf{Z}_i'\Sigma_{\epsilon i}^{-1}\mathbf{y}_i)\right],$$

is the best linear unbiased estimator (BLUE) of β (see [3, p. 233]). The remaining problem of applying (15) is that Ω is not known a priori and therefore must be estimated from the data. The next section examines various techniques for estimating Ω.

GENERALIZED LEAST SQUARES ESTIMATORS

Case 1: No Sampling Error

Consider first a special case where n_i approaches infinity for each choice situation i. This is a relatively common condition when $\boldsymbol{p}_i$ represents market share data for competing brands. When $\boldsymbol{p}_i$ is computed on the basis of thousands of units purchased, the sampling error term becomes negligible, and

$$\lim_{n_i\to\infty}\Sigma_{\epsilon i} = \Sigma_{\zeta i} = \sigma_\zeta^2\left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right).$$

In this case the GLS estimator (15) reduces to

$$\hat{\beta}_{gl} = \left[\sum_{i=1}^{M}(\mathbf{Z}_i'\Sigma_{\zeta i}^{-1}\mathbf{Z}_i)\right]^{-1}\left[\sum_{i=1}^{M}(\mathbf{Z}_i'\Sigma_{\zeta i}^{-1}\mathbf{y}_i)\right].$$

Since the rank of $\Sigma_{\zeta i}$ is only $m - 1$, it is necessary to delete one observation on choice objects from each choice situation i, thereby reducing $\Sigma_{\zeta i}$ to $(m - 1) \times (m - 1)$, before the matrix can be inverted. After deleting one object (say, the jth), $\hat{\beta}_{gl}$ can be written as:

$$(16) \qquad \hat{\beta}_{gl} = \left[\sum_{i=1}^{M}\mathbf{Z}_i^{(j)\prime}\frac{1}{\sigma_\zeta^2}\left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right)^{-1}\mathbf{Z}_i^{(j)}\right]^{-1}$$
$$\cdot\left[\sum_{i=1}^{M}\mathbf{Z}_i^{(j)\prime}\frac{1}{\sigma_\zeta^2}\left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right)^{-1}\mathbf{y}_i^{(j)}\right]$$
$$= \left[\sum_{i=1}^{M}\mathbf{Z}_i^{(j)\prime}(\mathbf{I}+\mathbf{J})\mathbf{Z}_i^{(j)}\right]^{-1}\left[\sum_{i=1}^{M}\mathbf{Z}_i^{(j)\prime}(\mathbf{I}+\mathbf{J})\mathbf{y}_i^{(j)}\right]$$

where $\mathbf{Z}_i^{(j)}$ and $\mathbf{y}_i^{(j)}$ are formed by deleting the jth row of $\mathbf{Z}_i$ and $\mathbf{y}_i$, respectively.[1] This result shows that it is not necessary to have prior knowledge of σ_ζ^2 in order to obtain $\hat{\beta}_{gl}$. Furthermore, McGuire, Farley, Lucas, and Ring [11, pp. 1207-8] proved that $\hat{\beta}_{gl}$ given by (16) is equivalent to the OLS estimator, $\hat{\beta}_o$. To prove this, first rearrange the rows of $\mathbf{Z}_i$ and $\mathbf{y}_i$ such that the deleted object becomes the last row of $\mathbf{Z}_i$ and $\mathbf{y}_i$. This is permissible because the choice of the object to be deleted may be arbitrary (see [11, p. 1205]). Then we may write [11, p. 1208]:

$$\mathbf{Z}_i = \begin{pmatrix} \mathbf{I} \\ \hdashline -\mathbf{1} \end{pmatrix} \mathbf{Z}_i^{(j)} \text{ and } \mathbf{y}_i = \begin{pmatrix} \mathbf{I} \\ \hdashline -\mathbf{1} \end{pmatrix} \mathbf{y}_i^{(j)},$$

where $\mathbf{I}$ is the $(m-1) \times (m-1)$ identity matrix and $\mathbf{1}$ is the $(m-1)$ element row vector of 1's. With this notation,

$$\begin{aligned}
\hat{\beta}_o = (\mathbf{Z}'\mathbf{Z})^{-1}(\mathbf{Z}'\mathbf{y}) &= \left[\sum_{i=1}^{M} \mathbf{Z}_i'\mathbf{Z}_i\right]^{-1}\left[\sum_{i=1}^{M} \mathbf{Z}_i'\mathbf{y}_i\right] \\
&= \left[\sum_{i=1}^{M} \mathbf{Z}_i^{(j)\prime}(\mathbf{I} \;\vdots\; -\mathbf{1}')\begin{pmatrix} \mathbf{I} \\ -\mathbf{1} \end{pmatrix}\mathbf{Z}_i^{(j)}\right]^{-1} \\
&\quad \cdot \left[\sum_{i=1}^{M} \mathbf{Z}_i^{(j)\prime}(\mathbf{I} \;\vdots\; -\mathbf{1}')\begin{pmatrix} \mathbf{I} \\ -\mathbf{1} \end{pmatrix}\mathbf{y}_i^{(j)}\right] \\
&= \left[\sum_{i=1}^{M} \mathbf{Z}_i^{(j)\prime}(\mathbf{I}+\mathbf{J})\mathbf{Z}_i^{(j)}\right]^{-1}\left[\sum_{i=1}^{M} \mathbf{Z}_i^{(j)\prime}(\mathbf{I}+\mathbf{J})\mathbf{y}_i^{(j)}\right] = \hat{\beta}_{gl},
\end{aligned}$$

since $\mathbf{1}' \cdot \mathbf{1} = \mathbf{J}$. This shows that, under the assumption of no sampling error, the OLS estimator, $\hat{\beta}_o$, is indeed the BLUE of β.

Case 2: No Specification Error

This special case assumes that there are no specification errors,[2] i.e.,

$$\Sigma_{\epsilon i} = \Sigma_{\eta i}.$$

If the sample size n_i is sufficiently large, we may approximate $\Sigma_{\eta i}$ by (12). In practice π_i is not known a priori, and hence $\Sigma_{\eta i}$ must be estimated from observed proportions $\mathbf{p}_i$. Let $\mathbf{P}_i$ be an $m \times m$ diagonal matrix with diagonal entries $(p_{i1}, p_{i2}, \ldots, p_{im})$, and estimate $\Sigma_{\eta i}$ by, using (12),

$$\hat{\Sigma}_{\eta i} = \frac{1}{n_i}\left[\mathbf{P}_i^{-1} - \frac{1}{m}(\mathbf{J}\mathbf{P}_i^{-1} + \mathbf{P}_i^{-1}\mathbf{J}) + \hat{c}\mathbf{J}\right] \tag{17}$$

where:

$$\hat{c} = \frac{\sum_{j=1}^{m}\left(\frac{1}{p_{ij}}\right)}{m^2}.$$

The GLS estimator of β in this case is given by

$$\hat{\beta}_{g2} = \left[\sum_{i=1}^{M}(\mathbf{Z}_i'\hat{\Sigma}_{\eta i}^{-1}\mathbf{Z}_i)\right]^{-1}\left[\sum_{i=1}^{M}(\mathbf{Z}_i'\hat{\Sigma}_{\eta i}^{-1}\mathbf{y}_i)\right]. \tag{18}$$

Since $\hat{\Sigma}_{\eta i}$ has rank $m-1$, it is necessary that one observation be dropped for each i before it can be inverted. It is recommended that the observation with the smallest p_{ij} be dropped.[3]

The exact statistical properties of the GLS estimator (18) are unknown because of approximations involved in the derivation of (12) and because $\mathbf{p}_i$ is used instead of π_i for obtaining $\hat{\Sigma}_{\eta i}$. In order to examine the properties of $\hat{\beta}_{g2}$, a Monte Carlo simulation was performed. The values of three independent variables, x_{kij}, were generated randomly for three choice situations and for six objects (i.e., $k = 3$, $M = 3$, and $m = 6$), then transformed into z_{kij}. The value of β was set at (.5, −.3, .8), and π_i ($i = 1, 2, 3$) were computed using (2). The Monte Carlo technique was used to generate $\mathbf{p}_i$ from π_i. Each run consisted of drawing n_i random numbers and allocating them to m categories based on the multinomial sampling assumption. Then $\mathbf{p}_i$ were computed and $\hat{\beta}_{g2}$ was estimated for each run. Also $\hat{\beta}_o$ was estimated for comparison purposes.[4] Three sample sizes of n_i = 50, 100, and 200 were tried. A total of 100 independent runs were made for each sample size. Table 1 summarizes the simulation results.

In comparing the GLS and OLS estimates with the true parameter values, two criteria were used. One is the variance of the estimates over 100 trials. Another is the mean square error defined as:

$$\text{MSE}(\hat{\beta}_k) = \text{E}(\hat{\beta}_k - \beta_k)^2.$$

The mean square error can be decomposed into two parts as follows:

$$\begin{aligned}
\text{MSE}(\hat{\beta}_k) &= \text{E}(\hat{\beta}_k - \bar{\hat{\beta}}_k)^2 + (\bar{\hat{\beta}}_k - \beta_k)^2 \\
&= \text{Var}(\hat{\beta}_k) + (\text{bias})^2,
\end{aligned}$$

where $\bar{\hat{\beta}}_k$ is the mean of estimates $\hat{\beta}_k$. Inspection of the table reveals the following points:

1. The GLS estimator $\hat{\beta}_{g2}$ has smaller variances than the OLS estimator $\hat{\beta}_o$, as expected.
2. $\hat{\beta}_{g2}$ underestimates the absolute values of parameters, especially for small sample sizes, but it is still a better

[1] Both I and J are $(m-1) \times (m-1)$.

[2] Although it is difficult to conceive of situations where this assumption is valid, we felt it necessary to consider this case since maximum likelihood approach of Haines et al. [4] is based on this assumption.

[3] This consideration would be unnecessary if $\Sigma_{\eta i}$ were the exact covariance matrix.

[4] $\hat{\beta}_{gl}$ is not applicable here because it only applies to the cases where sampling errors can be ignored.

Simulation Results[a]

	True value of parameters		
	β_1	β_2	β_3
	.5000	*−.3000*	*.8000*
	Estimated parameters		
Estimating conditions	$\hat{\beta}_1$	$\hat{\beta}_2$	$\hat{\beta}_3$
Sample Size = 50			
OLS estimates			
Mean	.4870	−.3124	.8274
Variance	.01828	.01932	.02085
Mean square error	.01845	.01948	.02160
GLS estimates			
Mean	.4707	−.2915	.7179
Variance	.01115	.01311	.01322
Mean square error	.01202	.01318	.01996
Sample Size = 100			
OLS estimates			
Mean	.5161	−.3063	.8254
Variance	.01541	.01288	.01065
Mean square error	.01567	.01292	.01130
GLS estimates			
Mean	.4801	−.2704	.7734
Variance	.00788	.00836	.00749
Mean square error	.00827	.00924	.00820
Sample Size = 200			
OLS estimates			
Mean	.5114	−.2984	.8144
Variance	.00573	.00731	.00457
Mean square error	.00586	.00731	.00478
GLS estimates			
Mean	.4987	−.2962	.7858
Variance	.00333	.00455	.00393
Mean square error	.00333	.00456	.00413

[a]Means, variances, and mean square errors of estimates are computed on the basis of 100 trials for each sample size.

estimator than $\hat{\beta}_o$ in terms of the mean square error criterion.

3. $\hat{\beta}_{g2}$ is apparently consistent. ($\hat{\beta}_o$ is known to be consistent.)

The tendency of the GLS estimator to underestimate the absolute values of parameters is a result of the fact that the approximation of the logarithm by only the linear term of the Taylor series expansion around π_i is not very good for small sample sizes. But, since the direction of the bias is known, its existence is not a critical problem. In hypothesis testing, the GLS estimates would probably give greater t-values than the OLS estimates when tested against the null hypothesis because the estimated variances of the parameters are smaller.

The simulation results are reassuring in two ways. The fact that $\hat{\beta}_{g2}$ was a better estimator than $\hat{\beta}_o$ (using the mean square error criterion) for n_i as small as 50 suggests that the estimated contemporaneous covariance matrix (17) is sufficiently accurate. This in turn suggests that the approximate covariance matrix (12) is a reasonable approximation even for such a small sample size. Furthermore, the use of p_i instead of π_i appears to be a reasonable procedure.

Case 3: Both Types of Error Present

We now turn to the case where both specification and sampling errors are present. Unlike two previous cases, it is not possible to estimate $\Sigma_{\epsilon i}$ a priori, primarily because one lacks the knowledge of σ_ζ^2. In this case, therefore, the following two-step procedure is suggested. First, use the OLS estimation technique and compute the residual term. If we let the results be $\hat{\epsilon}_o$, it is known that $\hat{\epsilon}_o$ is a consistent estimate of ϵ. Goldberger [3, p. 238] shows that:[5]

$$\begin{aligned}\mathbf{E}\hat{\epsilon}_o'\hat{\epsilon}_o &= \operatorname{tr}[\Omega - (\mathbf{Z}'\mathbf{Z})^{-1}(\mathbf{Z}'\Omega\mathbf{Z})] \\ &= \sum_{i=1}^{M} \operatorname{tr}(\Sigma_{\eta i} + \Sigma_{\zeta i}) \\ &\quad - \operatorname{tr}\left[(\mathbf{Z}'\mathbf{Z})^{-1}\left\{\sum_{i=1}^{M} \mathbf{Z}_i'(\Sigma_{\eta i} + \Sigma_{\zeta i})\mathbf{Z}_i\right\}\right].\end{aligned}$$

For a sufficiently large sample size ($n_i > 50$, say), we can estimate $\Sigma_{\eta i}$ by (17). Then we have:

$$\begin{aligned}\mathbf{E}\hat{\epsilon}_o'\hat{\epsilon}_o &= \sum_{i=1}^{M} \operatorname{tr}\left[\sigma_\zeta^2\left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right) + \hat{\Sigma}_{\eta i}\right] \\ &\quad - \operatorname{tr}\left[(\mathbf{Z}'\mathbf{Z})^{-1}\left(\sum_{i=1}^{M} \mathbf{Z}_i'\hat{\Sigma}_{\eta i}\mathbf{Z}_i\right)\right] \\ &\quad - \operatorname{tr}\left[(\mathbf{Z}'\mathbf{Z})^{-1}\left\{\sum_{i=1}^{M} \mathbf{Z}_i'\sigma_\zeta^2\left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right)\mathbf{Z}_i\right\}\right] \\ &= \sigma_\zeta^2 \operatorname{tr}\left[\sum_{i=1}^{M}\left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right)\right] - \operatorname{tr}\left[\sum_{i=1}^{M}\hat{\Sigma}_{\eta i}\right] \\ &\quad - \operatorname{tr}\left[(\mathbf{Z}'\mathbf{Z})^{-1}\left(\sum_{i=1}^{M} \mathbf{Z}_i'\hat{\Sigma}_{\eta i}\mathbf{Z}_i\right)\right] \\ &\quad - \sigma_\zeta^2 \operatorname{tr}\left[(\mathbf{Z}'\mathbf{Z})^{-1}\left\{\sum_{i=1}^{M}\mathbf{Z}_i'\left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right)\mathbf{Z}_i\right\}\right].\end{aligned}$$

If we let $Q = \hat{\epsilon}_o'\hat{\epsilon}_o$ be the estimated residual sum of squares, then σ_ζ^2 can be consistently estimated by:

$$\hat{\sigma}_\zeta^2 = \frac{Q + \operatorname{tr}\left[\sum_{i=1}^{M}\hat{\Sigma}_{\eta i}\right] + \operatorname{tr}\left[(\mathbf{Z}'\mathbf{Z})^{-1}\left(\sum_{i=1}^{M}\mathbf{Z}_i'\hat{\Sigma}_{\eta i}\mathbf{Z}_i\right)\right]}{M(m-1) - \operatorname{tr}\left[(\mathbf{Z}'\mathbf{Z})^{-1}\left\{\sum_{i=1}^{M}\mathbf{Z}_i'\left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right)\mathbf{Z}_i\right\}\right]}.$$

In the second step, $\Sigma_{\zeta i}$ is estimated by

$$\hat{\Sigma}_{\zeta i} = \hat{\sigma}_\zeta^2\left(\mathbf{I} - \frac{1}{m}\mathbf{J}\right),$$

and a GLS estimator of the following form is obtained:

[5]tr **A** stands for the trace of matrix **A**, i.e., the sum of the main diagonal elements of **A**.

$$\hat{\beta}_{g3} = \left[\sum_{i=1}^{M} \mathbf{Z}_i' (\hat{\Sigma}_{\eta i} + \hat{\Sigma}_{\zeta i})^{-1} \mathbf{Z}_i \right]^{-1} \cdot \left[\sum_{i=1}^{M} \mathbf{Z}_i' (\hat{\Sigma}_{\eta i} + \hat{\Sigma}_{\zeta i})^{-1} \mathbf{y}_i \right]. \tag{19}$$

As before, matrix $(\hat{\Sigma}_{\eta i} + \hat{\Sigma}_{\zeta i})$ has rank $m - 1$ and hence can be inverted only after one observation per choice situation is dropped.

The GLS estimator given by (19) is rather cumbersome to use because it requires special programming. In some situations the iterative generalized least squares (IGLS) procedure suggested in [2] may be more practical to use.[6] IGLS requires the assumption that the contemporaneous covariance matrix for the error vector ϵ_i is invariant for all choice situations, that is,

$$\Sigma_{\epsilon i} = \Sigma_{\epsilon} \tag{21}$$

for all i ($i = 1, 2, \ldots, M$). Given this assumption, the following two-step procedure may be used. In the first step, the OLS procedure is used to estimate ϵ. If we let σ_{jl} be the (j, l)th element of Σ_{ϵ}, σ_{jl} can be consistently estimated by:

$$\hat{\sigma}_{jl} = \frac{\sum_{i=1}^{M} \hat{\epsilon}_{ij} \hat{\epsilon}_{il}}{M - 1}$$

where $\hat{\epsilon}_{ij}$ is the OLS estimate of ϵ_{ij}. Then in the second step a GLS estimator of the form:

$$\hat{\beta}_{g4} = \left[\sum_{i=1}^{M} (\mathbf{Z}_i' \hat{\Sigma}_{\epsilon}^{-1} \mathbf{Z}_i) \right]^{-1} \left[\sum_{i=1}^{M} (\mathbf{Z}_i' \hat{\Sigma}_{\epsilon}^{-1} \mathbf{y}_i) \right] \tag{22}$$

is obtained. The IGLS estimate results if the above process is repeated until $\hat{\beta}_{g4}$ converges.

One advantage of IGLS over $\hat{\beta}_{g3}$ is that it can handle contemporaneous correlations among specification errors. So far we have assumed that specification errors are independently distributed, that is,

$$\mathrm{E}(\log \zeta_i^* \cdot \log \zeta_i^{*\prime}) = \sigma_{\zeta}^2 \mathbf{I} \tag{23}$$

where $\mathbf{I}$ is the $m \times m$ identity matrix. This assumption may in some cases be deemed unrealistic. A more general assumption is that:

$$\mathrm{E}(\log \zeta_i^* \cdot \log \zeta_i^{*\prime}) = \mathbf{V}_i$$

where $\mathbf{V}_i$ is an arbitrary covariance matrix. Unfortunately, there is no estimation technique developed under this general assumption. But if one further assumes that:

$$\mathbf{V}_i = \mathbf{V} \tag{24}$$

for all i and that there is no sampling error (see [2]), assumption (21) is satisfied and $\hat{\beta}_{g4}$ or the IGLS estimator is more appropriate than $\hat{\beta}_{g3}$.

However, it must be pointed out that the IGLS assumption (21) is not always justified. More specifically, it is necessary that the data satisfy the following conditions:

1. The same set of m choice objects are observed in all choice situations. All observed proportions are greater than zero for at least $m - 1$ objects.[7]
2. If sampling errors are not negligible, observed proportions p_i are reasonably constant over i.

If condition 1 is violated, the pair of objects designated by subcript set (i,j) and (i,k) may not be the same as another pair designated by set (i', j) and (i', k) $(i' \neq i)$, and hence there is no a priori basis to assume $E(\log \zeta_{ij}^* \cdot \log \zeta_{ik}^*) =$ constant for all i. The violation of condition 2 would result in the fluctuation of $\Sigma_{\eta i}$ over i, which in turn causes $\Sigma_{\epsilon i}$ to vary. It may be noted that $\hat{\beta}_{g1} (= \hat{\beta}_o)$, $\hat{\beta}_{g2}$, and $\hat{\beta}_{g3}$ do not depend on these conditions on the data set and therefore are more flexible in their application. To be sure, this flexibility is the result of the simplifying assumption (23), but in most situations where assumption (24) is not justified (that is, if condition 1 is not satisfied), the assumption of uncorrelated specification errors would probably be the most natural one to use.

CONCLUSION

In this article we have presented parameter estimation techniques for the Multiplicative Competitive Interaction model. The potential usefulness of the model to marketing practitioners and researchers, we believe, justifies the development of estimation techniques which can be easily implemented. All least squares estimators, with the exception of $\hat{\beta}_{g3}$, require little more than usual multiple regression analysis programs. In particular, if sampling errors are negligible (as in many marketing applications where the dependent variable is market share figures) and specification errors are uncorrelated, simple transformations would permit one to use OLS estimation programs. Now that practical estimation techniques are available, it is hoped that more uses will be made of the MCI model in marketing research and planning.

APPENDIX

In order to derive the approximate covariance matrix of ϵ_{ij}, first define:

[6] Beckwith [1] was probably the first to apply IGLS (or IZEF for Zellner Iterative Aitken Estimator) in the marketing context, but Bultez and Naert were the first to apply IGLS to estimate the parameters of the MCI model.

[7] This condition is necessary because any choice object with $p_{ij} = 0$ must be excluded from the data for choice situation i. This causes the number of objects per choice situation, m, to vary over i.

$$\pi_i' = (\pi_{i1}\ \pi_{i2}\ \pi_{i3} \dots \pi_{im})$$
$$p_i' = (p_{i1}\ p_{i2} \dots p_{im})$$
$$\log \pi_i' = (\log \pi_{i1} \log \pi_{i2} \dots \log \pi_{im})$$
$$\log p_i' = (\log p_{i1} \log p_{i2} \dots \log p_{im}).$$

The rationale for the following approximate covariance matrix is the fact that, for a reasonably large sample size, the random vector p_i is mostly concentrated around its expected value π_i. Since the y_i variables are defined in terms of $\log p_i$, we examine the Taylor series expansion of $\log p_i$ around $p_i = \pi_i$ to obtain an approximate linear transformation of p_i into $\log p_i$. Since $\log p_{ij}$ is a univariate function in p_{ij} despite the constraint that $\Sigma_j p_{ij} = 1$, the Taylor series expansion of $\log p_{ij}$ around π_{ij} is given by:

$$\log p_{ij} = \log \pi_{ij} + \left.\frac{d \log p_{ij}}{dp_{ij}}\right|_{p_{ij}=\pi_{ij}} (p_{ij} - \pi_{ij}) + R_{ij}$$
$$= \log \pi_{ij} + \frac{1}{\pi_{ij}}(p_{ij} - \pi_{ij}) + R_{ij}$$

where R_{ij} is the remainder term. In a matrix notation,

$$\log p_i = \log \pi_i + \Pi_i^{-1}(p_i - \pi_i) + R_i \tag{25}$$

where Π_i is an $m \times m$ diagonal matrix with diagonal entries of $(\pi_{i1}\ \pi_{i2} \dots \pi_{im})$ and R_i is the vector of the remainder terms.

Recalling the definition of $y_{ij} = (\log p_{ij}/\tilde{p}_i)$, then $y_i = K \log p_i$, where K is the $m \times m$ normalization matrix of the form:

$$K = \begin{pmatrix} \left(1 - \frac{1}{m}\right) & -\frac{1}{m} & \dots & -\frac{1}{m} \\ -\frac{1}{m} & \left(1 - \frac{1}{m}\right) & \dots & -\frac{1}{m} \\ \dots & \dots & \dots & \dots \\ -\frac{1}{m} & -\frac{1}{m} & \dots & \left(1 - \frac{1}{m}\right) \end{pmatrix}$$
$$= I - \frac{1}{m} J.$$

where I is the $m \times m$ identity matrix, and J is the $m \times m$ summation matrix (all elements equal unity). We may then write, using (25),

$$y_i = K \log \pi_i + K \Pi_i^{-1}(p_i - \pi_i) + K R_i.$$

For a reasonably large sample size, p_i is approximately normally distributed with mean vector π_i and covariance matrix:

$$\Sigma_{pi} = \frac{1}{n_i} \begin{pmatrix} [\pi_{i1}(1 - \pi_{i1})] & -\pi_{i1}\pi_{i2} & \dots & -\pi_{i1}\pi_{im} \\ -\pi_{i2}\pi_{i1} & [\pi_{i2}(1 - \pi_{i2})] & \dots & -\pi_{i2}\pi_{im} \\ \dots & \dots & \dots & \dots \\ -\pi_{im}\pi_{il} & -\pi_{im}\pi_{i2} & \dots & [\pi_{im}(1 - \pi_{im})] \end{pmatrix}$$
$$= \frac{1}{n_i}(\Pi_i - \Pi_i J \Pi_i),$$

where n_i is the sample size for choice situation i.

Ignoring the residual term in the Taylor series expansion, we may write:

$$\eta_i \cong y_i - K \log \pi_i = K \Pi_i^{-1}(p_i - \pi_i).$$

For a sufficiently large sample size for each i, η_i is approximately normally distributed with mean vector:

$$E\eta_i = K \Pi_i^{-1}(E p_i - \pi_i) = 0,$$

and contemporaneous covariance matrix:

$$\Sigma_{\eta i} = \Sigma_{yi} = \mathrm{Var}[K \Pi_i^{-1} p_i]$$
$$= K \Pi_i^{-1} \Sigma_{pi} \Pi_i^{-1\prime} K'$$
$$= \left(I - \frac{1}{m}J\right) \Pi_i^{-1} \frac{1}{n_i}(\Pi_i - \Pi_i J \Pi_i) \Pi_i^{-1} \left(I - \frac{1}{m}J\right)$$
$$= \frac{1}{n_i}\left[\Pi_i^{-1} - \frac{1}{m}(J \Pi_i^{-1} + \Pi_i^{-1} J) + cJ\right]$$

where $c = \Sigma_{j=1}^{m} (1/\pi_{ij})/m^2$.

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